

Фазовые переходы

В предельных случаях:

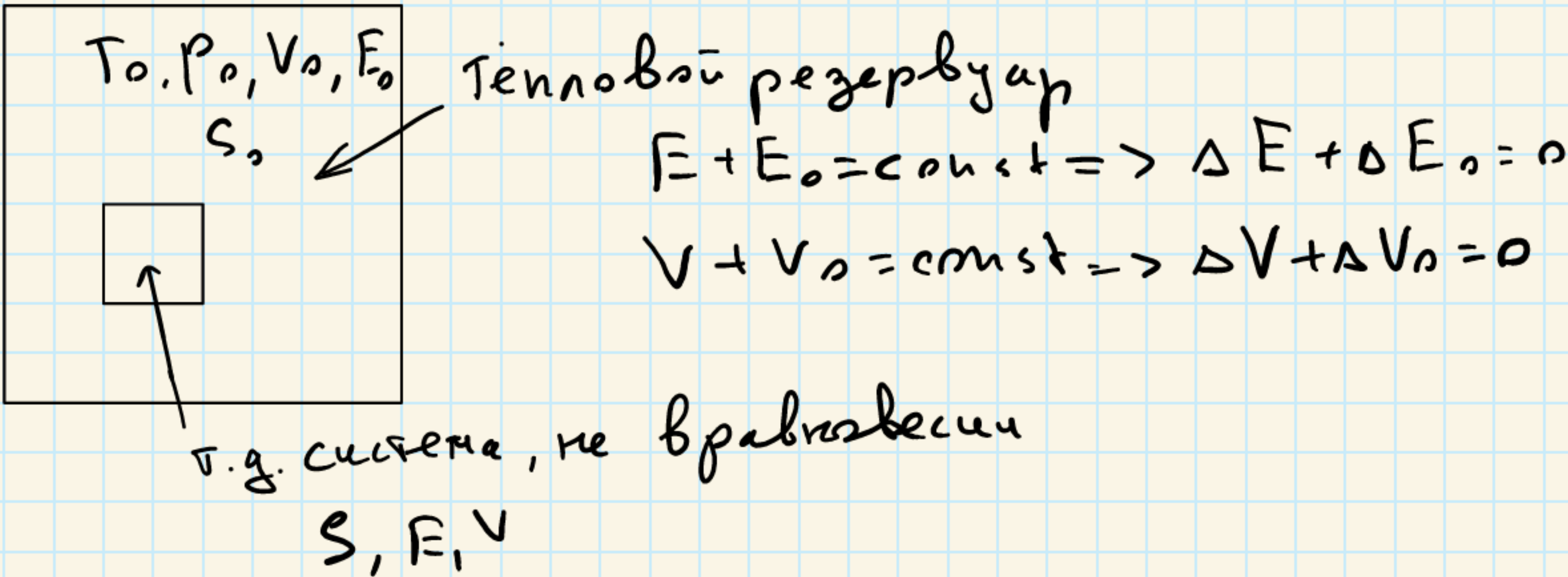
$$\Gamma_{nol} = \Pi \cdot \sigma$$

$$S_{nol} = - \Pi \frac{d\sigma}{dT}$$

$$U_{nol} = \Gamma_{nol} + T S_{nol} = \Pi \left(\sigma - T \frac{d\sigma}{dT} \right)$$

Условия равновесия

- $P_1 = P_2$
- $T_1 = T_2$
- Равновесие отн. перехода частиц. $\Phi_1 = \Phi_2$



$$\Delta F = T_0 \Delta S_0 - P_0 \Delta V_0$$

$$\Delta S_0 = \frac{\Delta E_0}{T_0} + \frac{P_0}{T_0} \Delta V_0$$

$$\Delta S_{tot} = \Delta S_0 + \Delta S = \frac{\Delta E_0}{T_0} + \frac{P_0}{T_0} \Delta V_0 + \Delta S \geq 0$$

$$\Delta E_0 + P_0 \Delta V_0 + T_0 \Delta S \geq 0$$

$$-\Delta E - P_0 \Delta V + T_0 \Delta S \geq 0$$

$$\Delta E + P_0 \Delta V - T_0 \Delta S \leq 0$$

$$\Delta (E + P_0 V - T_0 S) \leq 0$$

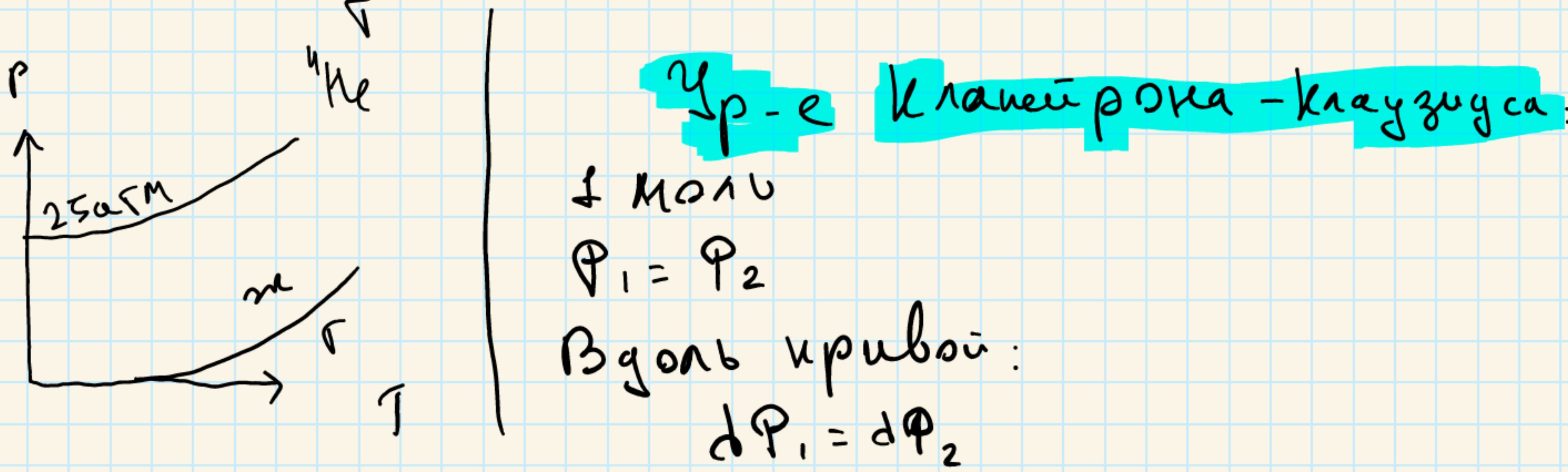
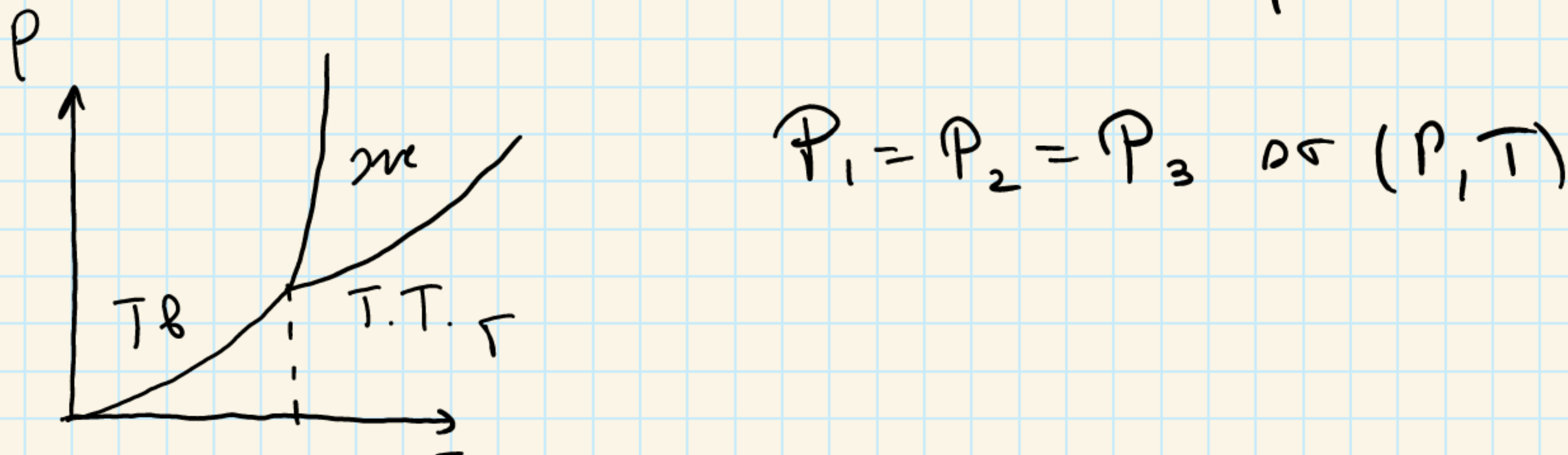
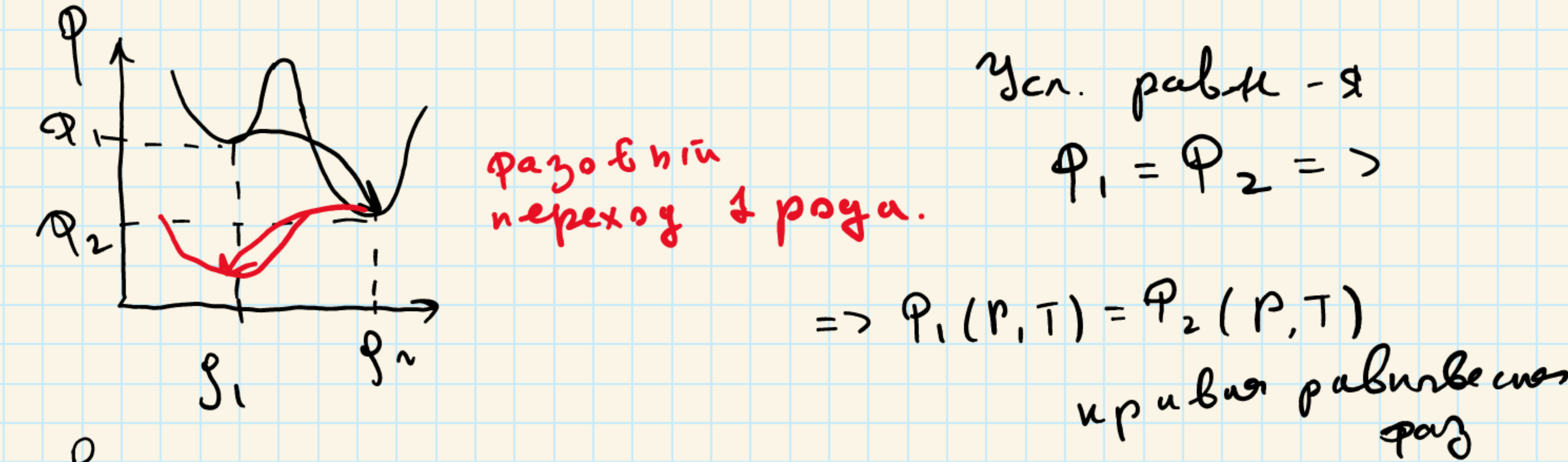
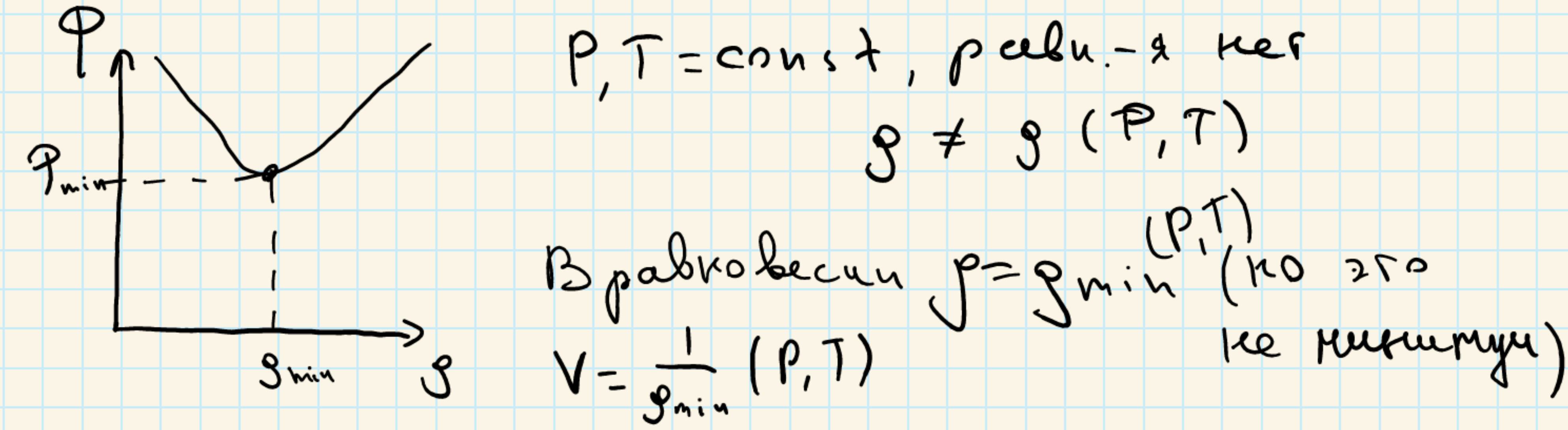
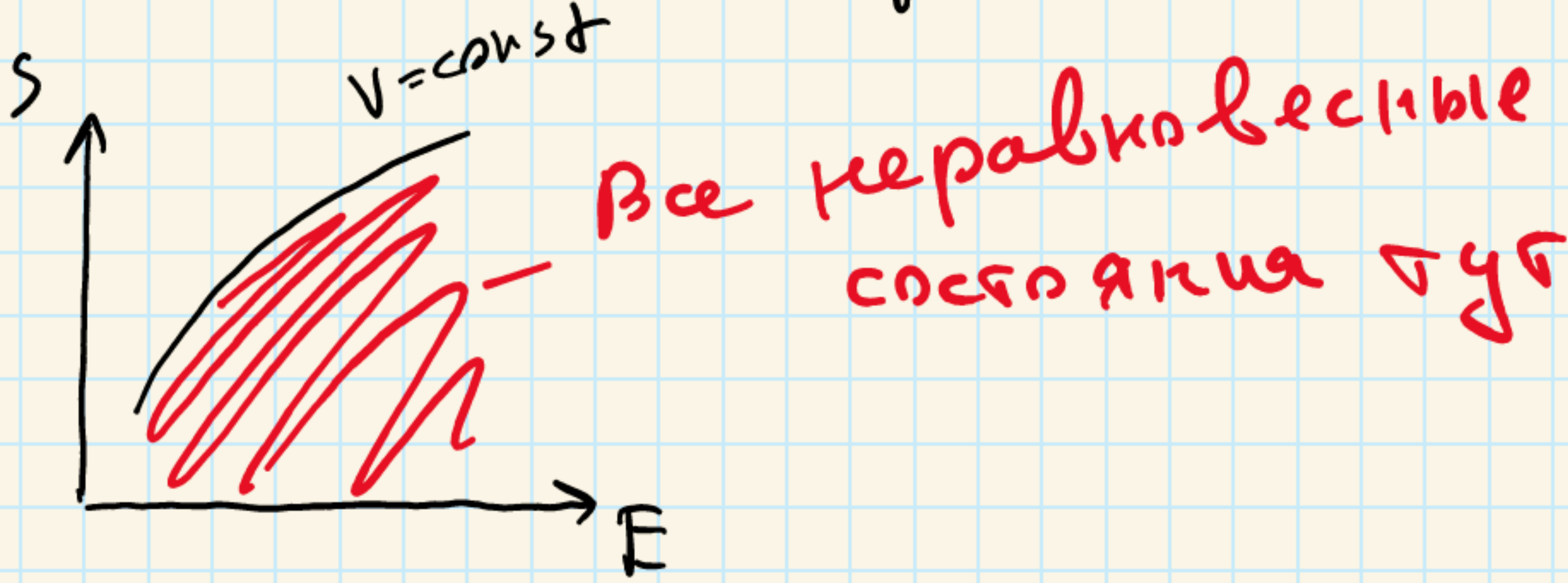
$$\Delta \Phi \leq 0$$

В равновесии $\Phi = \min (P_0, T_0 = \text{const})$

Замечание: А если $\Delta V = 0$

$$\Delta (E - T_0 S) \leq 0, \Rightarrow F \leq 0$$

$$F = \min \text{ в равновесии.}$$



$$-S_1 dT + v_1 dP = -S_2 dT + v_2 dP$$

$$\frac{dP}{dT} = \frac{S_2 - S_1}{-v_1 + v_2}$$

$$q = T(S_2 - S_1)$$

$$\frac{dP}{dT} = \frac{q}{T(v_2 - v_1)}$$

ур-е Клапейрона-Клаузиуса.

Для перехода жидкости $\Rightarrow$ пар $v_2 \gg v_1, \frac{v_2}{v_1} \sim 10^3$

$$\frac{dP}{dT} \approx \frac{q}{T v_2} = \frac{q P}{R T^2} \Rightarrow \frac{dP}{P} = \frac{q}{R} \frac{dT}{T^2}$$

$$\ln \frac{P}{P_0} = \frac{q}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right)$$

$$P = P_0 e^{\frac{q}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right)}$$